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Operationalizing AI for Crisis & Emergency Management

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Abstract

Crisis and Emergency Management (CEM) in high-risk sectors such as oil & gas, petrochemicals, and transportation faces growing challenges from operational complexity, inter-agency coordination gaps, and evolving hazards. While compliance frameworks such as ICS, OPITO MEMIR, and ISO 22320 set essential standards, many organizations still struggle to maintain continuous operational readiness. This paper explores how Artificial Intelligence (AI), cognitive models, and integrated digital tools can operationalize readiness — transforming it from periodic drills into a continuous, measurable capability.

Through AI-driven predictive risk modeling, adaptive simulation, performance benchmarking, and real-time incident management, organizations can significantly reduce mobilization time, improve decision-making accuracy, and enhance compliance resilience. Case applications from offshore platforms, port facilities, and national disaster agencies demonstrate quantifiable benefits, including reduced communication delays, optimized resource allocation, and improved inter-agency coordination.

The discussion concludes with a forward-looking roadmap, showing how AI will evolve from an enhancement to an essential element of CEM over the next five years. The results offer a compelling business and operational case for early adoption — positioning AI not as a future option, but as the foundation for safer, smarter, and more resilient crisis management.

Keywords: Emergency response, training, simulator, artificial intelligence, large language models

Introduction

Why Crisis & Emergency Management (CEM) Must Evolve

Global industry trends, recent high-impact incidents, and increasing regulatory scrutiny underscore the need for a fundamental shift in how organizations prepare for, respond to, and recover from emergencies. While compliance with established standards remains critical, operational agility and adaptive capacity now determine whether response efforts succeed or fail. This paper examines the drivers behind this shift and the role of AI in creating a continuous readiness model that bridges the gap between policy and practice.

Industry Incident Landscape

A special interest group of IChemE Safety & Loss Prevention, developed a report, "Learning Lessons from Major Incidents". In this report, root causes of various incidents since Piper Alpha in 1988, were analysed to identify underlying lessons. The lessons learnt in crisis and emergency from one prominent incident from each sector is given in Table below:

Table 1—Crisis & Emergency Incident Summary

Sector	Year	Event Name	Country	Type	Fatalities / Injuries	CEM Lessons
Upstream	2015	Camarupim	Brazil	Explosion	09	Compliance / Training: No scenario-specific procedure for confirmed unignited gas leaks; reliance on basic memory cards without detailed instructions. Competence: Misapplied detection protocols indicate training and role clarity gaps. Response: Ad-hoc decision-making led to personnel entering a confirmed hazardous area, increasing exposure risk.
Ship	1989	Valdez	USA	Pollution	0	Compliance / Planning: No adequate vessel traffic monitoring; violation of navigation procedures; insufficient spill response contingency planning. Training / Competency: Inadequate training, poor fatigue management, and unqualified officer on watch at a critical time. Response: Remote location hindered timely emergency response; inadequate spill response equipment and communication delays increased environmental impact.
Rail	2013	Lac Megantic	Canada	Fire	47	Compliance / Planning: No effective emergency response plan for major spills in populated areas; inadequate regulatory oversight of railway safety practices. Response: Lack of mutual aid preparedness and coordination for large-scale fire/explosion; communication and strategy gaps delayed effective incident control.
Downstream	2012	Richmond	USA	Fire	26	Compliance / Planning: No leak response protocol; Incident Command lacked necessary technical expertise to identify hazard cause. Training / Competency: Miscommunication of process conditions (actual temperature ~640°F) led to unsafe actions; Stop Work Authority ineffective due to cultural barriers. Response: Underestimated hazard zones; unsafe mitigation actions (clamping corroded pipe) exposed personnel to vapor cloud.
Terminal	2005	Buncefield	UK	Explosion	0	Compliance / Planning: Inadequate emergency planning for major spills and multi-tank fires; containment systems not designed for combined rainwater and fuel runoff. Training / Competency: Gaps in emergency response readiness due to human factors (staff under pressure, inadequate familiarity with emergency procedures). Response: Limited capacity to manage large-scale fire and environmental runoff; bund and drainage systems not capable of preventing spread to sensitive areas.

From the above table it is seen that the gaps in crisis and emergency management often lies in lack of compliance, inadequate planning, training and competency issues and response. The IChemE report was further analyzed to identify the root causes over 25 incidents starting in 1966 to 2015. The various incidents analyzed and given in below supported by a bar chart summarizing the root causes.

Incidents Analysed:

Piper Alpha	Longford Gas Plant	Grangemouth	Texas City
Mumbai High North	Exxon Valdez	Milford Haven	Delaware City
Macondo / Deepwater Horizon	Lac Megantic	Avon	McKee
Cidade de Sago Mateus FPSO	Feyzin	Izmit	Anacortes
Skikda LNG	Romeoville	Humber	Chiba
Richmond	Torrance	San Juan Ixhuatepec	Buncefield
Bayamon			

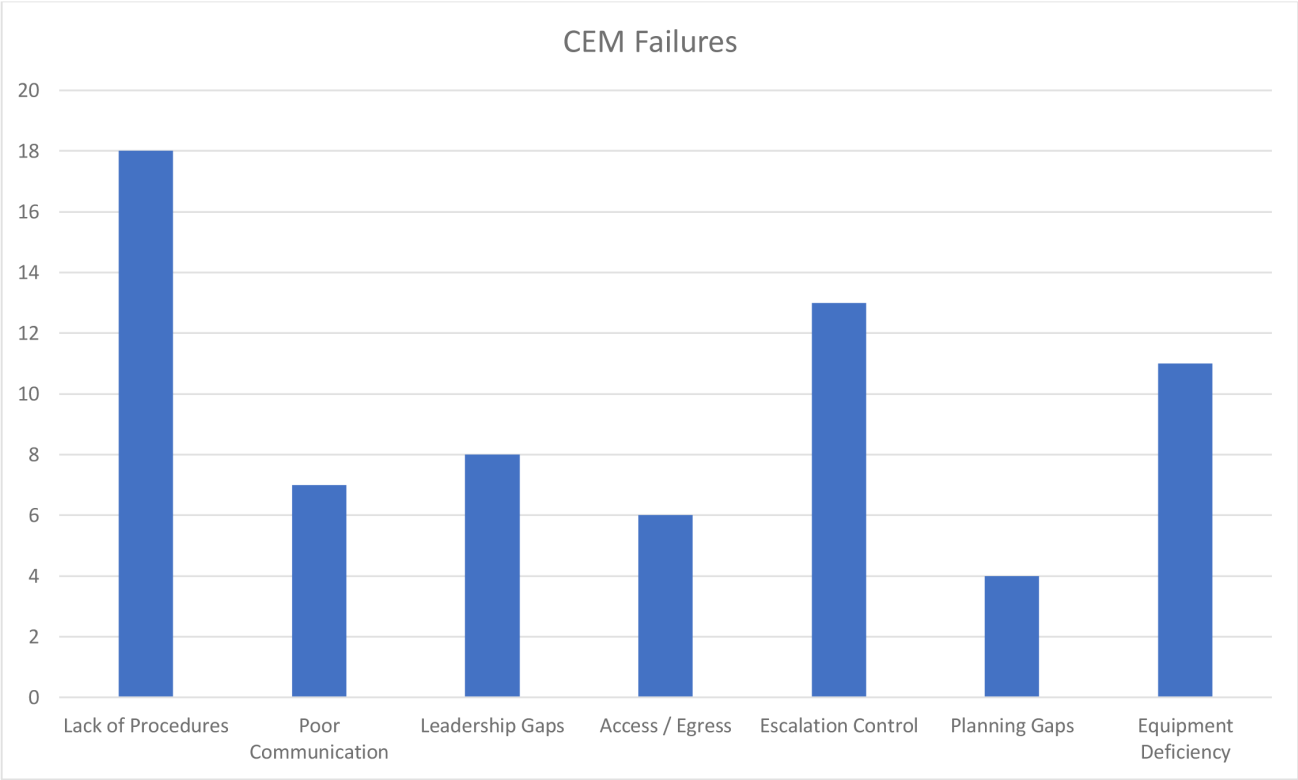


Figure 1—CEM Failures in Incidents

Reviewing the CEM failures chart, it is clear that the most prevalent weakness across oil & gas incidents is a **lack of procedures**, present in the majority of cases. This indicates a widespread gap in scenario-specific emergency protocols and formalized response frameworks. **Escalation control** and **equipment deficiencies** also feature prominently, reflecting vulnerabilities in preventing incident spread and ensuring critical systems remain functional during crises. Leadership gaps, poor communication, and access/egress issues are less frequent but still significant, often compounding the primary failures. Overall, the data underscores that while technical safeguards are important, procedural readiness and operational discipline remain the most critical determinants of effective crisis and emergency management.

The incident analyses demonstrate that conventional Crisis & Emergency Management (CEM) approaches — characterized by static plans and procedures, periodic drills, and post-event reporting — are inadequate for addressing the dynamic, high-consequence scenarios encountered in the oil and gas sector. Operational environments are increasingly complex, involving multi-hazard risks, interdependent assets, and geographically distributed teams. These realities necessitate the integration of advanced, data-driven technologies capable of augmenting human decision-making and improving the fidelity of situational awareness.

Large Language Models (LLMs) have demonstrated capability in rapid information synthesis, procedural guidance retrieval, and multilingual communication support. Integrated into operational workflows, LLMs can ingest multi-source data — risk analysis, COMAH reports, SCE failures, field reports, and other regulatory documentation — and generate relevant and probable crisis and emergency scenarios for the team to prepare and practice.

Cognitive modelling frameworks, such as the Adaptive Control of Thought-Rational (ACT-R) architecture, offer a means of quantitatively assessing human performance under operational stress. By simulating the perception-decision-action cycle, these models can be used to identify bottlenecks in information processing, diagnose skill attrition over time, and design training interventions that target specific cognitive deficits observed in team performance data.

Virtual Emergency Operations Centres (EOCs), leveraging secure, cloud-based collaboration platforms, enable distributed teams to share a unified operational picture. When integrated with LLMs and asset-specific digital twins, these EOCs can function as adaptive coordination hubs, supporting both onshore and offshore incident management in real time, as well as serving as high-fidelity environments for simulation-based training. This dual-use capability allows organizations to employ the same platform for preparedness exercises and live incident coordination, ensuring continuity between training scenarios and operational execution.

The convergence of these technologies establishes a foundation for **predictive, adaptive, and performance-informed CEM**. Rather than focusing solely on documenting outcomes, technologically enhanced CEM systems can continuously anticipate evolving hazards, recommend optimized courses of action, and evaluate execution effectiveness. This represents a substantive shift from reactive postures toward **proactive, resilience-oriented operational readiness** in high-risk industrial domains.

Business Case for AI in Crisis & Emergency Management

The integration of Artificial Intelligence into Crisis & Emergency Management (CEM) is no longer a speculative proposition but a demonstrable pathway to measurable operational, financial, and regulatory benefits. In high-risk sectors such as oil and gas, where asset integrity, human safety, and reputational capital are tightly interlinked, AI-enabled systems offer a compelling value proposition that aligns with both corporate performance goals and broader sustainability commitments.

From a **regulatory and compliance** perspective, AI can support continuous adherence to industry frameworks and international standards. Automated documentation, performance auditing, and real-time readiness scoring provide verifiable evidence for regulators and insurers, reducing exposure to penalties and enabling favourable risk assessments.

Financial impact is equally tangible. Case studies indicate that AI-enabled anomaly detection and predictive resource allocation can reduce mobilization times by up to 40%, cut training logistics costs by 30-60%, and prevent incident escalation that could otherwise result in multimillion-dollar losses. Loss prevention extends beyond direct asset damage, safeguarding production continuity and minimizing unplanned downtime.

In terms of **readiness and resilience**, AI-based cognitive modeling and performance analytics identify capability gaps before they manifest in live operations. This allows targeted interventions, enhancing overall team efficiency and improving readiness indices by measurable margins. The same platforms that manage live incidents can also deliver high-fidelity simulation training, ensuring skills are continuously reinforced.

Importantly, AI adoption in CEM supports the **United Nations Sustainable Development Goals (SDGs)**, particularly SDG 9 (Industry, Innovation, and Infrastructure), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). These alignments are not merely corporate social responsibility statements; they translate into operational benefits such as reduced environmental impact

through optimized emergency containment, improved community safety, and enhanced resilience against climate-related hazards.

By framing AI deployment in CEM as both a **performance multiplier** and a **risk mitigation strategy**, decision-makers can justify budget allocation with a clear return on investment. This positions AI not as an experimental add-on, but as an essential component of a modern, resilient, and regulation-aligned crisis management capability.

Translating the Business Case into Applied Enhancements with ALERTSim

Procedural Completeness:

- ALERTSim supports procedural completeness by dynamically generating SOPs. This is enabled by the AI Scenario Creator, which simulates new and evolving operational scenarios.

For instance, when a refinery integrates a new hydrocracker unit, ALERTSim automatically generates SOPs for potential fire, toxic release, or power outage scenarios—providing procedural completeness even for operations not yet captured in legacy manuals.

Decision Authority:

- ALERTSim embeds role clarity and authority matrices into its Virtual EOC, reducing delays in critical decision-making. The platform further customizes the Emergency Response Plan (ERP) structure to reflect each user organization's hierarchy, ensuring alignment with international frameworks, national regulations, and company-specific standards.

For example, an offshore oil and gas operator can mirror its existing ERP chain of command within ALERTSim while simultaneously mapping roles to ICS or OPITO MEMIR frameworks. This dual alignment ensures that responders operate within familiar company structures, while maintaining compliance and interoperability with external regulators and mutual aid partners.

Communication Protocols:

- ALERTSim integrates digital radio, structured notification ladders, and standardized message templates to ensure consistent communication.

For example, during a port chemical spill exercise, ALERTSim triggers tiered notifications from first responders up to the Emergency Operations Centre, ensuring standardized language and escalation timing across agencies.

Training & Simulation:

- AI-driven simulations and performance analytics ensure scalable, continuous readiness.

For example, a national disaster agency can simulate cascading flood and industrial spill scenarios, with ALERTSim tracking individual and team performance metrics. The results are then benchmarked against compliance standards, creating a cycle of targeted re-training and sustained operational readiness.

Change Management:

- ALERTSim facilitates structured adaptation by embedding change control into emergency processes. This ensures updates to procedures, communication protocols, or authority matrices are seamlessly integrated across the organization.

For example, when a facility adds a new crude oil storage tank, ALERTSim automatically integrates the change into the ERP structure, updates response checklists to include the new asset, and ensures compliance standards are upheld during the transition.

The applied enhancements are further categorized into the 5 pillars essential for a Crisis and Emergency Management Platform. The [figure 2](#) below illustrates the ALERTSim Pillars.



Figure 2—ALERTSim Pillars illustrating the essentials of a Crisis and Emergency Management Platform

Scalability & Integration:

- A critical consideration for large operators is how emergency management platforms interface with existing enterprise systems and scale across distributed operations. ALERTSim is designed with integration at its core. Through secure APIs and modular connectors, it links seamlessly with ERP systems such as SAP or Oracle, SCADA/DCS environments, and ICS- based emergency protocols. This ensures that asset data, alarms, and procedural updates are exchanged, avoiding the silos that often hinder coordinated response.

Beyond technical integration, ALERTSim is architected for scalability. Its federated Virtual EOC capability allows multiple sites to operate autonomously during local incidents while remaining connected to a centralized oversight structure. This enables consistent adherence to international standards while allowing for local adaptation.

AI for Live Emergency Response Teams and Simulation (ALERTSim)

The application of Artificial Intelligence (AI) to live emergency response teams and simulation-based training addresses the entire lifecycle of emergency readiness — from prevention and anticipatory action to live incident management. This section outlines the research foundation, conceptual framework, and applied case studies demonstrating operational transformation enabled by AI.

Research Behind the Project

The project builds upon interdisciplinary research integrating AI, cognitive modeling, and emergency management frameworks such as ICS and MEMIR. Data from historical incidents, simulation exercises, and live operations has been used to develop predictive models, performance benchmarks, and scenario generation engines. These models have been validated through controlled trials and real-world exercises.

ALERTSim is designed around the principle of human-machine teaming, ensuring that AI augments rather than replaces decisionmakers in high-stress environments. By automating repetitive and data-intensive tasks—such as scenario generation, real-time log analysis, and After-Action Report drafting—the platform reduces cognitive load and mitigates decision fatigue among responders. At the same time, human operators remain central to validation and contextual judgment, reinforcing accountability in critical moments. To build trust in AI outputs, ALERTSim emphasizes transparency through audit trails, explainable recommendations, and benchmarking against established standards. This balance of efficiency and interpretability enables responders to trust but verify AI guidance, resulting in faster, more confident, and better-coordinated crisis management actions.

Concepts & Framework

The framework employs AI-driven modules for:

- **Prevention & Anticipatory Action:** Predictive analytics for hazard detection, enabling proactive mitigation before escalation.
- **Response Planning & Strategic Readiness:** Decision-support algorithms that optimize resource allocation and coordination across multi-site operations.
- **Simulation Exercising:** High-fidelity, scenario-based training environments replicating operational stressors, communication constraints, and resource limitations.
- **Quantifying Readiness:** Cognitive performance metrics and AI-based scoring to identify skill degradation and prioritize training interventions.
- **Live Incident Management:** Integration of AI analytics, digital twins, and Virtual EOCs to deliver real-time situational awareness and adaptive response strategies.

Figure 3 illustrates ALERTSim AI + Human collaboration closed-loop framework. The cycle begins with AI-driven review of Emergency Response Plans against international / company specific standards, followed by automated gap analysis, scenario generation, simulation, and creation of an After Action Report.

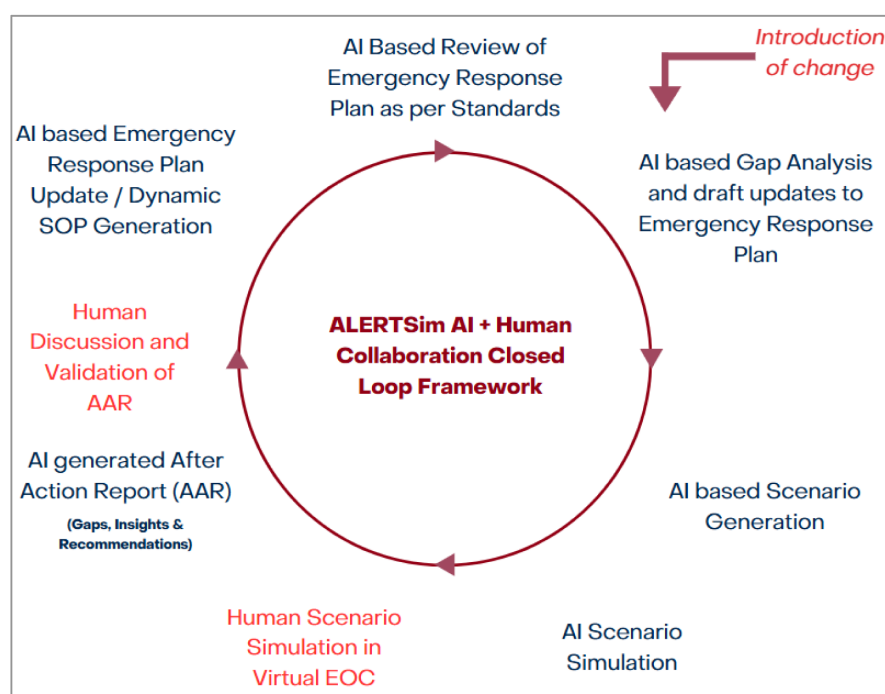


Figure 3—ALERTSim AI + Human Collaboration Closed-loop framework

Human teams then validate these AI outputs, conduct scenario-based testing in the Virtual EOC, and refine the ERP with contextual insights. This model reduces manual workload significantly, as most procedural and analytical steps are performed by AI, while human expertise strengthens decision-making, validation, and readiness through testing. The collaboration ensures both efficiency gains and reinforced trust in emergency preparedness.

AI Based Application Features

The AI based application modules based on the above framework is summarized in the table below:

Table 2—AI Application Modules

Sr. No.	Framework	Application Modules
1	Prevention & Anticipatory Action	1. AI Based Gap Analysis Reviews a. International and Company standards b. Hazard & Risk Analysis c. Analysis of changes to operations, hazard profiles, roles and responsibilities etc. d. Historical data, hazard trends, lessons learnt
2	Response Planning & Strategic Readiness	1. Review of Resource Allocation 2. Review of coordination protocols 3. Review of Mutual Aid Agreements
3	Simulation Exercising	1. AI Scenario Creator a. Multi-hazard scenarios b. Dynamic scenario generation c. Escalations & Failures 2. Virtual Emergency Operations Centre a. Simulated information boards b. Simulated communication channels c. Customizable Forms d. Multiple Command Rooms for ICS structures e. Back Room for Simulation observers and evaluators 3. AI Assisted After Action Reviews
4	Quantifying Readiness	1. AI + Cognitive Model Based Performance Evaluation 2. Scenarios can be evaluated through AI simulations and compared to Human Simulations 3. AI Performance Reports with Readiness Benchmarking
5	Live Incident Management	1. Virtual Emergency Operations Centre 2. Mobile Application for Onsite Personnel 3. Audit Ready with Recorded Sessions and Tracking

For executive audiences, the Virtual Emergency Operations Centre (EOC) is shown in [Figure 4](#). The dashboard integrates incident logs, communication channels, and KPI panels into a unified view, enabling faster decision-making under stress. Further information about the application architecture and features can be presented upon request.

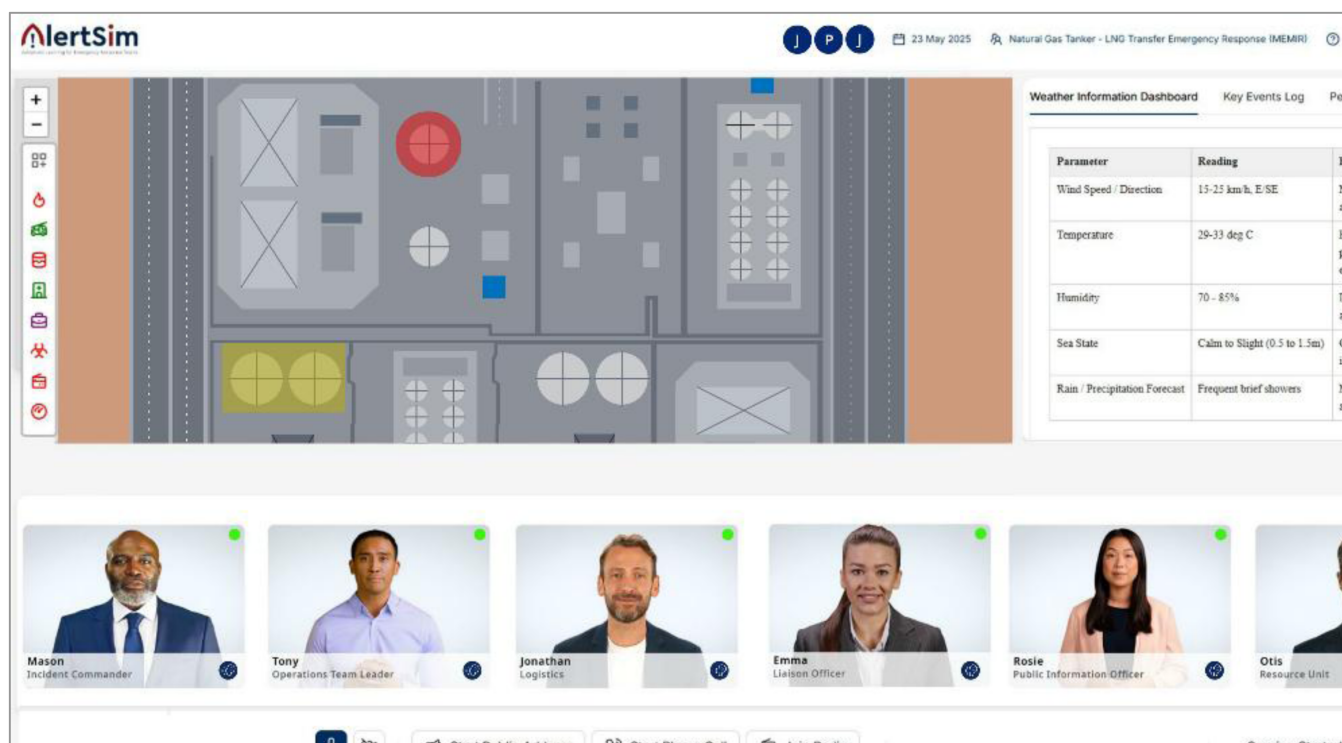


Figure 4—ALERTSim Virtual EOC

Case Study: OceanLNG Crisis and Emergency Management

This case study demonstrates how the ALERTSim AI framework enhances Compliance, Planning, and Simulation/Exercise (SimEx) for crisis and emergency management in a coastal LNG import and regasification terminal (OceanLNG). The process applied mirrors a realistic engineering lifecycle:

1. establish a standards-aligned Emergency Response Plan (ERP)
2. assess its completeness against a change in operations
3. validate proposed improvements through a tabletop exercise
4. and close the loop by updating documents, training, and metrics.

Major-accident hazard (MAH) facilities must ensure that emergency preparedness keeps pace with operational change. Traditional document reviews are time-consuming, subjective, and difficult to sustain as procedures evolve. ALERTSim addresses these issues with an AI-assisted pipeline that ingests ERP/SOP content, maps it to regulatory and industry standards, detects gaps, and generates role-specific instructions that are validated in exercises and fed back into controlled documents

OceanLNG Terminals Ltd., headquartered at the Harbour Industrial Zone in Coastal City, operates a coastal, shallow-water berth LNG import and regasification terminal. Our operations encompass the reception of LNG from carriers, its safe transfer to insulated storage tanks, regasification through high-pressure vaporizers, and delivery into the national pipeline network. The company's mission is to maintain the highest operational standards in safety, environmental stewardship, and service reliability.

The site is strategically positioned along a sheltered coastal bay with a shallow-water berth specifically designed to accommodate Q-Flex and Q-Max LNG carriers. The terminal serves as a critical energy infrastructure node for the national gas grid, handling the reception, storage, regasification, and onward transmission of LNG to domestic and industrial consumers.

The company's core operational components include:

- **Marine Facilities:** A 400-meter jetty with cryogenic loading arms and mooring dolphins equipped with quick-release hooks.
- **Storage Facilities:** Two full-containment cryogenic LNG storage tanks, each with a capacity of 180,000 m³, equipped with in-tank pumps and boil-off gas management systems.
- **Regasification Systems:** High-pressure open rack vaporizers (ORVs) with seawater intake and discharge facilities, supplemented by submerged combustion vaporizers (SCVs) for peak demand periods.
- **Control and Safety Systems:** Distributed Control System (DCS) with Safety Instrumented Systems (SIS), emergency shutdown systems (ESD), and an integrated fire and gas detection network.

The company maintains an Emergency Response Plan that provides structured guidance for responding to incidents that could harm personnel, assets, the environment or company reputation. It translates regulatory and industry best practice requirements from the HSE COMAH regulations and offshore major accident hazards framework into actionable procedures tailored to LNG operations.

This case study demonstrates how AI enabled application for crisis and emergency management can be deployed to maintain the procedures, training and simulation. The example considered in the case study is of a plant modification at OceanLNG and its impact on emergency response management.

ALERTSim is deployed to:

- **Assess procedure completeness and standards alignment**
- **(Identify risk-response gaps** by detecting missing clauses, ambiguous instructions, and gaps with regards to a defined change.
- **Generate actionable, role-specific instructions** (action cards/SOP drafts) and validate them through tabletop simulation, creating an auditable evidence trail for updates.

The table below defines the case study input parameters:

Table 3—Case Study Input Parameters

Sr. No.	Parameter	Description
1	Asset :	OceanLNG import & regasification terminal with shallow-water berth
2	Document Reviewed :	ERP-LNG-001 v1.0 with appendices (equipment, training/drills, incident reporting, media templates, compliance register).
3	Event demanding change in ERP :	Berth upgrade to enable side-by-side FSRU operations with ship-to-ship (STS) LNG transfer and SIMOPS with shore.
4	Standards Complied To :	COMAH-aligned guidance, NFPA 59A, ISO 28460, ISO 22320/22361, IGC/SIGTTO good practice

Data and Assumptions

1. The ERP reflects an LNG terminal without prior Ship-To-Ship capability;
2. Floating, Storage and Regasification Unit (FSRU) introduction represents a material change to hazards, roles, communications, and marine coordination.
3. Training, equipment registers, and media templates are available and version-controlled.
4. Exercise outcomes and KPI targets are defined before SimEx and captured on standardized logs for auditability.

AI Routines Deployed

Several AI routines are available in the application for reviewing documentation, scenario creation, validation of simulations and for closing the loop by updating the ERP documentation. The routines used in the case study are listed in the table below:

Table 4—AI Routines

Sr. No.	Parameter	Description
1	Ingestion & Graph Mapping :	The ERP/SOP corpus is ingested; clauses are normalized and mapped into a standards knowledge graph. Cross-references (roles, equipment IDs, forms, appendices) are resolved to a canonical ontology
2	Metrics :	ALERTSim computes Standards Coverage Index (SCI, presence of required clauses per standard/topic) and Procedure Completeness Score (PCS clarity/operability of role-specific steps, triggers, verifications). Results are reported per section/role and visualized as a heat map.
3	Gap Reasoning :	The system flags missing (not covered), ambiguous (non-actionable), and broken (invalid references) items and proposes candidate redlines.
4	Actionable Drafts :	For high-risk gaps, ALERTSim generates draft SOPs and condensed action cards (e.g., "Abort STS & Secure Transfer") tied to ICS roles and measurable KPIs.
5	SimEx Validation	A tabletop exercise tests decision ladders and timings (ESD-1/2, ERS, exclusion zones, Port/VTS coordination, public information), capturing structured data on forms.
6	Closed-Loop Update	Findings are converted into controlled redlines for the ERP, SOPs, training calendar, and maps, with re-scoring (SCI/PCS) to quantify readiness uplift.

Key Outputs

The key outputs from the AI application included the following:

1. Baseline SCI/PCS for the new scope (FSRU/STS) indicating moderate standards coverage but low procedural completeness.
2. Gap analysis proposing additions to
 - a. Facility/Ops description,
 - b. Communication annexure,
 - c. Incident Classification triggers,
 - d. Ship To Ship authority matrix,
 - e. Equipment addendum,
 - f. Training scenarios/KPIs,
 - g. Public Information variants, and
 - h. map requirements.
3. Tabletop AAR with quantified timings against targets and root-cause analysis of delays.
4. Updated ERP/SOP drafts and an operationalization roadmap (Virtual EOC, micro-drills, re-score cadence).

The change event that Demands an update of ERP

OceanLNG will enable side-by-side operations between a Floating Storage and Regasification Unit (FSRU) and an LNG carrier (LNGC) to conduct ship-to-ship (STS) LNG transfer at the shallow-water berth while certain shore systems remain energized (SIMOPS). The change introduces a second hull, dual command structures, and additional safety instrumented interfaces (ESD-1/2) across ship-ship and ship-shore boundaries.

The upgrade increases send-out flexibility and berth utilization but materially alters the hazard/operability envelope: moving interfaces, mooring dynamics, hose/arm emergency release (ERS) sequencing, exclusion zones, and multi-agency coordination with Port/VTS.

Operating envelope:

- **Sea state & wind:** allowable significant wave height (Hs) and wind speeds to be specified by marine engineering study; the ERP shall encode red/amber/green thresholds and abort triggers.
- **Visibility & current:** minimum visibility for approach/transfer; cross-current limits for tugs and station-keeping.
- **Systems dependencies:** verified ESD-link integrity, cargo control interlocks, firewater/foam availability, and communications redundancy as pre-conditions to commence transfer.

SIMOPS boundaries:

The ERP will define what shore activities may continue during STS, which are curtailed, and how work permits, access control, and hot-work restrictions adapt to the dual-hull configuration.

AI Gap Analysis

The AI gap analysis routine analysed the coverage of the emergency response plan for the proposed change by determining a Standards Coverage Index (SCI) and Procedure Completeness Score (PCS). The outcome generated was SCI for STS-specific topics is low, and the PCS is constrained by missing triggers, role authority, and acceptance criteria.

Indicative baseline scores (FSRU/STS topics)

- **SCI:** ~15% across key sections (facility/ops, hazards, comms, classification, roles, equipment, training).
- **PCS:** near-zero in most sections; partial signal only where generic marine controls exist.

Gaps were categorized as **Missing** (no clause present), **Ambiguous** (present but non-actionable), and **Broken** (invalid or absent cross-reference). Representative examples are shown below.

Table 5—AI Gap Analysis

ERP Area	Gap Category	Example
Facility/Ops Description	Missing	No FSRU asset profile, STS overview, SIMOPS boundaries, or metocean thresholds.
Marine Interface (Comms/ESD)	Missing/Ambiguous	No ESD-1/2 test & abort protocol; no plain-language PAGA or radio phrases for "Stop cargo/Abort STS."
Incident Classification	Missing	No STS-specific triggers (link fault timers, drift/tension excursions, ERS activation).
Roles & Responsibilities	Ambiguous	Authority matrix not defined for IC, FSRU Master, LNGC Master, STS PIC, Marine Coordinator.
Equipment & Deployment (Appx 4)	Missing	No ERS spares, ESD-test kit, mooring load monitoring, quick-release controls.
Training & Drills (Appx 6)	Missing	No STS scenarios/KPIs (ESD/ERS timing, degraded comms, night operations).
Maps (Appx 1)	Broken	Exclusion zones, dual-hull muster/egress, tug standby positions not yet issued.
Public Information	Ambiguous	Templates exist but lack Port/VTs co-sign variants for marine STS incidents.

A detailed log of required changes referencing the documents were generated by ALERTSim AI and saved for future reference. The AI went on to update the existing ERP to address all the gaps identified. A

robust document control system was established for document and revision control and was followed by the AI application. The detailed logs and updated ERP will be provided as appendices in the final manuscript.

Simulation Exercise (Tabletop)

ALERTSim AI is developed to design AI generated scenarios based on the company information, asset information, hazard profiles and emergency response documentation. The AI scenario creator provides a detailed Scenario Design document based on:

1. The proposed change and its impact on the layout and hazards
2. Its impact on the emergency response plans, resources, procedures etc.
3. Escalations and system failures (Not used for the case study, as the validity of the updated procedure was to be checked)

The AI scenario creator provided a detailed Simulation Design Document that provided a Master Sequence Event Lists, Systems to be Tested, Participant Lists and Inject information. These supporting documents will be included in the appendices in the final manuscript. The simulation exercise was performed in ALERTSim Virtual Emergency Operations Centre. The AI transcribed all the discussions from the simulation and mapped it to a timeline. This formed the basis for identifying further gaps in the procedures and training. The AI generated an After-Action Report (AAR) for the participants and observers to review. The simulation performance was measured against KPIs as presented in the table below:

Table 6—Simulation Key Performance Indicators

Metric	Target (Mins)	Achieved (Mins)	Delta (Mins)	Assessment
ESD go/no-go	≤ 5:00	9:10	+4:10	SOP absent; unclear abort authority
"Stop cargo" announced	≤ 0:45	1:40	+0:55	No PAGA script; talk group churn
"Manifolds closed" confirmed	≤ 2:30	4:20	+1:50	Split responsibilities not codified
"Drain complete" confirmed	≤ 8:00	14:30	+6:30	No controlled drain checklist
Exclusion zone established	≤ 4:00	8:05	+4:05	Criteria/phrases not predefined
Port/VTs notified	≤ 2:00	6:10	+4:10	Notification ladder missing
PIO first statement	≤ 15:00	25:00	+10:00	Joint wording negotiated ad-hoc
Comms continuity	≥ 99%	96%	-3%	Temporary radio overload
Muster complete	≤ 12:00	15:00	+3:00	Dual-hull muster map absent
ICS logs completeness	≥ 95%	85%	-10%	Peak-period under-logging

The following insights were also generated:

- **Decision latency cluster:** Mentions of "confirm/verify/stand-by" spiked when authority was unclear map to **Missing Authority Matrix**.
- **Phrase ambiguity:** Varied language for the same command (e.g., "hold transfer", "cease cargo", "stop pumps") add **plain-language PAGA & radio lexicon**.
- **Broken references:** Multiple calls to "see map" with no Appendix-1 issued **Broken** class; prioritize map production for dual-hull muster/egress and exclusion zones.

- **Log bottleneck:** Time-stamping lag associated with a single scribe **process control** fix: dual scribes (ICS log vs timeline) + Virtual EOC templates.

Followed by identifying the root cause of the above insights a Root-cause synopsis is produced: No STS-specific SOPs, no authority/decision ladder, incomplete comms annex, and absent maps were primary contributors to misses across the KPI set.

This was used as a basis by the AI to generate a recommendation list for updating the emergency management system. The recommendations are produced below:

Document changes (ERP v1.1):

- **FSRU/STS Overview:** SIMOPS boundaries, metocean thresholds, pre-conditions to start transfer.
- **Annex 8-A STS Comms Plan:** talkgroups/hierarchy, failover, PAGA scripts ("Stop cargo", "Abort STS"), horn/whistle equivalence, radio check cadence.
- **STS Triggers:** link-fault timers, LEL thresholds, pressure/flow excursions, ERS activation, mooring tension/drift exceedance, red weather lines.
- **Authority Matrix (STS):** decision ladder for IC, FSRU Master, LNGC Master, STS PIC, Marine Coordinator.
- **Equipment Addendum:** ERS spares, ESD-test kits, manifold blinds, hose drain/flare connections, mooring load monitoring, spare fenders, QRH controls.
- **Training & KPIs:** STS injects; timing targets; dual-scribe requirement; Virtual EOC templates.
- **Media Variants:** STS incident templates co-signed with Port/VTs.
- **Appendix 1 (Maps):** STS berth arrangement; exclusion zones; dual-hull muster/egress; tug standby; stowage.

The above changes were made to the emergency response plan by the AI.

Re-score and Readiness Validation

A further validation exercise was performed by the AI to rescore SCI and PCS:

1. SCI uplift (STS topics): From ~15%baseline to **≥75-85%** after ERP v1.1 inserts (coverage of comms, triggers, roles, equipment, training, maps).
2. PCs uplift (STS procedures): From ~15-20% to **≥70%** after issuing SOP-STs-001/002/003 and action cards. Residual gaps tied to real-world integration testing with vessel crews.

To quantify the impact of AI-assisted updates, KPI timings before and after ERP revisions were compared. [Figure 3](#) illustrates the improvements, demonstrating measurable reductions in response delays and increased standards coverage. KPI improvements from baseline to ERP v1.1 — showing reduced response times and uplift in Standards Coverage Index (SCI) and Procedure Completeness Score (PCS).

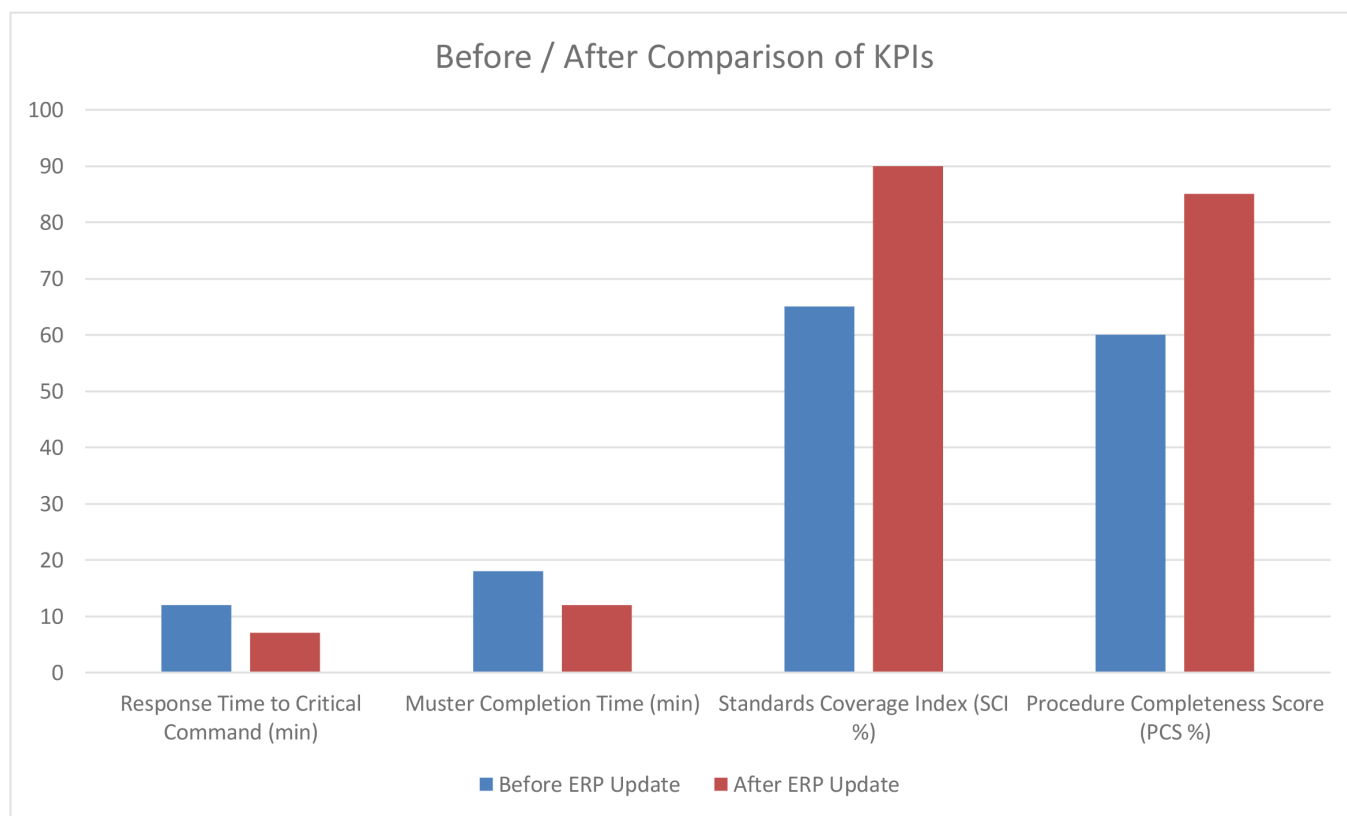


Figure 5—KPI Comparison before and after AI assisted updates

Table 7—Validation of Updated Emergency Response Plans and Procedures using SCI and PCS

ERP Area	SCI Baseline	SCI v1.1	PCS Baseline	PCS v1.1
Facility/Ops Description	0%	80%	0%	60%
Marine Interface (Comms/ ESD)	9%	85%	0%	75%
Incident Classification	0%	80%	0%	70%
Roles & Responsibilities	12%	90%	0%	75%
Equipment & Deployment	0%	70%	0%	60%
Training & Drills	43%	85%	0%	80%
Maps & Overlays	—	90%	—	70%

Impact of using ALERTSim AI for Emergency Response Management

The impact of using ALERTSim AI was determined qualitatively for the OceanLNG case study as below:
Indicative benefits at OceanLNG (qualitative).

- Time-to-safe: codified authority ladder and PAGA scripts reduce command latency in STS scenarios.
- Audit readiness: clause-to-evidence traceability shortens regulatory queries and third-party audit cycles.
- Training efficiency: reusable inject packs/action cards reduce exercise design and briefing effort.
- Reputation resilience: joint, pre-approved Port/VTs statements curb rumor propagation and restore normal operations faster.

ALERTSim AI's contribution is quantified along four dimensions:

- 1. loss avoidance through faster control and clearer decision authority,
- 2. compliance risk reduction via measurable clause coverage (SCI) and operability (PCS),
- 3. productivity gains in document maintenance and training preparation, and
- 4. stakeholder trust through consistent, bilingual public information.

Additional Case Studies across Various Sectors

To demonstrate the cross-industry applicability of the ALERTSim framework, it is important to look beyond oil and gas and illustrate how the same principles support resilience in other critical sectors. While the detailed OceanLNG case study highlights depth within energy operations, the following use cases provide breadth. Each example summarizes the operational context, a representative challenge—including emerging and black swan scenarios—and the way ALERTSim's AI-enabled modules address these gaps. This illustrates the generalizability of the approach, strengthening its relevance to utilities, transport, healthcare, and aviation infrastructure.

Table 8—Cross-industry applicability of ALERTSim framework

Sector	Context	Challenge	ALERTSim Application
Utilities (Power Grid)	National electricity grid operator responsible for millions of households and industries.	A cascading blackout triggered by cyberattack on SCADA systems (black swan). Limited visibility of incident escalation and fragmented communication between control centers.	ALERTSim runs AI-based gap analysis on ERP, generates SOPs for cyber-physical failures, and integrates Virtual EOC dashboards across regions. Ensures standardized communication protocols and decision authority for rapid restoration.
Transport (Metro Rail)	High-density metro network transporting over 3 million passengers daily.	Tunnel fire and toxic smoke release during peak hours leading to chaotic evacuation and lack of real-time passenger information.	AI-driven scenario simulation tests evacuation under varying conditions. ALERTSim SOP generator prepares dynamic passenger safety protocols. Performance tracking identifies weaknesses in muster/evacuation procedures for rapid improvement.
Medical (Hospitals Network)	Multi-hospital system in a coastal city exposed to natural disasters.	Mass-casualty surge following simultaneous hurricane impact and chemical spill (compound black swan). Strain on ERPs with unclear triage coordination across facilities.	ALERTSim generates updated ERP for compound emergencies, integrates triage authority matrices, and simulates patient routing between hospitals. Readiness scores highlight surge capacity gaps and guide resource allocation.
Airports (International Hub)	Major international airport with 80 million annual passengers.	Drone swarm attack disrupting runways and communication systems (emerging black swan threat). Lack of standardized cross-agency response between airport ops, ATC, police, and military.	ALERTSim Virtual EOC synchronizes agencies, standardizes language via structured notification ladders, and validates SOPs through scenario drills. AI simulations benchmark readiness against ICAO standards.

These cross-sector examples reaffirm that ALERTSim is not limited to energy operations but is adaptable to diverse infrastructures where readiness, coordination, and resilience are paramount. By addressing both conventional risks and black swan events, the framework demonstrates its role as a universally applicable enabler of modern crisis and emergency management. This broad applicability further strengthens the business case for AI in CEM, showing that the same investment not only delivers operational, financial, and compliance benefits within oil and gas, but also extends measurable value to utilities, transport, healthcare,

and aviation sectors. In doing so, ALERTSim positions itself as a cross-industry solution aligned with global standards and regulatory expectations.

Road Map for Operationalizing ALERTSim Platform

ALERTSim runs as a controlled service integrated with a document management system for versioned ERP/SOP updates, a Virtual EOC collaboration space with pre-built customizable logs and timelines, and a secure evidence store linking AARs, KPI traces, and test records to clauses.

Change Management

- Management of Change (MoC): each operational change (e.g., FSRU/STS) triggers an ALERTSim run → Gap Analysis → SME review → SimEx → v1.x issuance with tracked changes.
- Training calendar: codify monthly micro-drills (comms and timing) and quarterly table-tops; one annual full-scale with vessel crews (including night operations).
- Readiness dashboard: SCI/PCS trends, KPI pass rates, corrective action closure, and audit status reported quarterly to leadership

Interoperability: In the case of OceanLNG, the following interoperabilities were considered

- Marine partners: Port templates, talkgroup schemas, and notification ladders standardized and versioned.
- Mutual aid: resource catalogs and staging plans referenced from the compliance matrix; joint drills scheduled as per agreements.
- Multilingual releases: Arabic/English templates embedded in the PIO toolkit with synchronized approvals.

Model Risk Management and Ethics:

Decision Making:

1. All AI outputs (gap analysis, AI scenarios, action cards) are recommendations.
2. Subject-matter experts approve, modify, or reject; approvals are logged.
3. Decision authority in emergencies remains human.

Data Governance:

1. Classification: exercise versus live-incident data are tagged; personal data (medical logs) handled under privacy policy with minimum necessary fields.
2. Security: role-based access; encryption at rest and in transit; audit trails on every clause and edit.
3. Bias and validity: terminology normalization reduces unequal treatment of equivalent phrases; performance monitored across incident types and shifts.

Cyber Security Considerations

As digitalization advances, cybersecurity has become an equally critical pillar of crisis and emergency management. AI-driven Virtual EOCs and cloud-based simulations offer immense capability but also increase system complexity and, if unmanaged, can expand the attack surface of critical infrastructure.

1. ALERTSim addresses these risks by operating within a zero-trust security framework, applying end-to-end encryption, and enforcing strict role-based access controls.
2. Integration with corporate cybersecurity systems ensures alignment with established standards such as NIST and ISO 27001, while comprehensive audit trails safeguard both data integrity and accountability.

3. By embedding cybersecurity as a design principle rather than an afterthought, ALERTSim ensures that the resilience it provides in emergency management is not undermined by new digital vulnerabilities.

Limitations and External Validity

1. Document dependence: accuracy depends on ERP/SOP currency; absent artifacts (for example, maps) constrain validation.
2. Scenario fidelity: tabletop insights may not fully represent real ship motions, tug behavior, or plume dynamics.
3. Metric proxying: initial SCI/PCS rely on keyword/structure proxies prior to semantic calibration from multiple exercises.

Conclusions

ALERTSim operationalizes a repeatable, measurable pathway from compliance text to tested response procedures. In the OceanLNG case, introducing FSRU/STS revealed low baseline coverage for the new scope; AI-assisted redlines and SimEx closed priority gaps in command authority, communications, and marine controls. The framework links KPI improvements to avoided loss and ESG outcomes, while maintaining strong human oversight and auditability. Future iterations will incorporate live-incident telemetry, digital-twin coupling for plume and mooring dynamics, and automated re-scoring after each drill to sustain readiness.

Ongoing Further Developments - Crisis Management Simulations

Extend ALERTSim from operational emergency response into **strategic crisis management** simulations that stress governance, multi-stakeholder coordination, and executive decision-making under uncertainty. The focus is on ISO 22361-aligned crisis practice, reputation and legal exposure, and enterprise-level recovery.

Data-Driven Scenario Generation (Crisis Focus)

A generative engine fuses historical crises, hazard and incident databases, macro-geopolitical indicators, supply-chain telemetry, weather/ocean data, social-media signals, regulatory alerts, and company risk registers to produce **crisis narratives** with explicit causal chains. Outputs include evolving timelines, decision inflection points, stakeholder reactions, and resource/bottleneck constraints across jurisdictions.

Customizable Inputs (Governance & Risk Posture)

Participants configure scenarios by specifying:

- **Context:** geography, sector, asset portfolio, cross-border exposure.
- **Threat vectors:** safety events, cyber compromise, disinformation, activism, labor disruption, extreme weather, or multi-hazard cascades.
- **Governance parameters:** escalation thresholds, **decision authorities** (CMT/Board), disclosure obligations, risk appetite.
- **Resource posture:** internal teams, mutual aid, advisors (legal/PR), third-party dependencies. ALERTSim adapts complexity and inject cadence to learning objectives and team maturity.

Multi-Phase Crisis Modeling

Crisis progression is modelled across **sense-making** → **strategy setting** → **coordinated response** → **stabilization** → **recovery/claims/investigation**. Dynamic injects arise from state and performance, not

fixed timestamps (e.g., regulator directive, data leak, adverse media cycle, sudden weather shift), preserving realism while enabling targeted assessment.

Role-Specific Paths & Coordination

Information flows and decision rights are tailored to **Crisis Management Team (CMT)** roles and partner agencies: Crisis Manager, Operations, Safety, Legal/Compliance, Corporate Affairs/PIO, HR/People, Finance/Insurance/Claims, Supply Chain, IT/Cyber, Board Liaison; external: regulators, port authority, police, health/environmental agencies. Role views include controlled **information asymmetry**, authority ladders, and approval gates to rehearse coordination and strategic alignment.

Compliance Alignment & Auditability

Scenarios are tagged to **ISO 22361/22320**, sector guidance (e.g., IOGP, API RP 75), disclosure/notification rules (e.g., market disclosure, data protection), and internal policies. After-Action Reports are generated with clause-level links, decision logs, timestamps, and artifacts, ready for audit and **lessons-learned ingestion** back into policy and training.

Interactive Visualization & Executive Simulation

The platform renders synchronized artifacts for executive use: high-level **crisis dashboards**, stakeholder heat-maps, sentiment streams, decision timelines, and resource/status boards, with drill-downs to evidence.

Measurement & Learning

Key measures extend beyond operational KPIs to crisis-level performance:

- **Time to strategic intent** (first clear crisis statement); **time to regulator notification**.
- **Decision coherence index** (alignment of actions with policy and risk appetite).
- **Rumour half-life** and misinformation correction effectiveness.
- **Stakeholder sentiment delta** (pre/post intervention) and **coordination latency** across agencies. Results feed continuous improvement, with re-scoring of policy completeness and readiness.

Research Plan & Validation

Planned evaluations include A/B comparisons against traditional table-tops (time-to-criterion, retention), SME realism ratings, and inter-rater reliability of scoring. Cross-industry pilots will test transferability; bias and drift monitoring will be reported as part of model risk management.